

● MEDIUM

● ALGEBRA

Algebra

8 questions · ~12 min

19

Q1

ALGEBRA

For the linear function g , the graph of $y = g(x)$ in the xy -plane has a slope of 2 and passes through the point 1, 14. Which equation defines g ?

- A. $g(x) = 2x$
- B. $g(x) = 2x + 2$
- C. $g(x) = 2x + 12$
- D. $g(x) = 2x + 14$

Scientists collected fallen acorns that each housed a colony of the ant species *P. ohioensis* and analyzed each colony's structure. For any of these colonies, if the colony has x worker ants, the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y , in the colony. If one of these colonies has 58 worker ants, which of the following is closest to the predicted number of larvae in the colony?

- A. 41
- B. 61
- C. 83
- D. 190

$$y = 18 - 5x$$

The equation above represents the speed y , in feet per second, of Sheila's bicycle x seconds after she applied the brakes at the end of a ride. If the equation is graphed in the xy -plane, which of the following is the best interpretation of the x -coordinate of the line's x -intercept in the context of the problem?

- A. The speed of Sheila's bicycle, in feet per second, before Sheila applied the brakes
- B. The number of feet per second the speed of Sheila's bicycle decreased each second after Sheila applied the brakes
- C. The number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop
- D. The number of feet Sheila's bicycle traveled from the time she began applying the brakes until the bicycle came to a complete stop

Q4

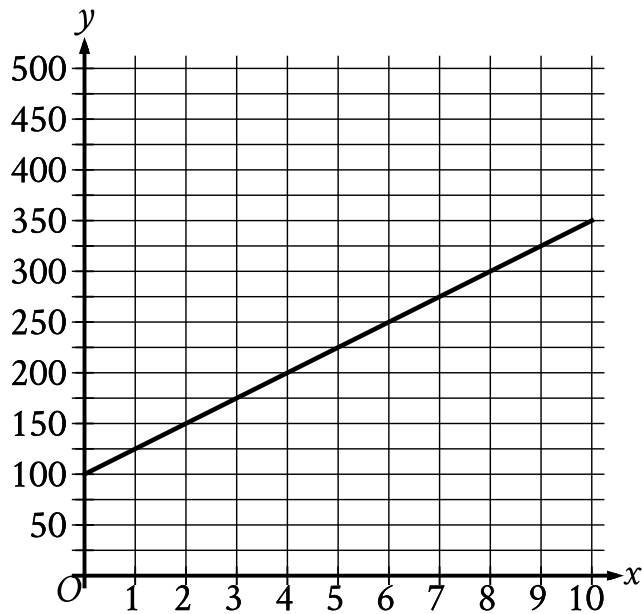
GRID-IN ALGEBRA

The function f is defined by $fx = -9x + 9$. What is the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



The graph of the function f , where $y = f(x)$, gives the total cost y , in dollars, for a certain video game system and x games. What is the best interpretation of the slope of the graph in this context?

- A. Each game costs \$25.
- B. The video game system costs \$100.
- C. The video game system costs \$25.
- D. Each game costs \$100.

x	$f(x)$
0	-2
2	4
6	16

Some values of the linear function f are shown in the table above. What is the value of $f(3)$?

- A. 6
- B. 7
- C. 8
- D. 9

$$f(x) = 39$$

For the given linear function f , which table gives three values of x and their corresponding values of fx ?

A.

x	fx
0	0
1	0
2	0

B.

x	fx
0	39
1	39
2	39

C.

x	fx
0	0
1	39
2	78

D.

x	fx
0	39
1	0
2	-39

In the xy -plane, the graph of the linear function f contains the points 0, 3 and 7, 31.
Which equation defines f , where $y = fx$?

A. $fx = 28x + 34$

B. $fx = 3x + 38$

C. $fx = 4x + 3$

D. $fx = 7x + 3$